

BIOSENSING PLATFORM & ML CLASSIFICATION SYSTEM
Technical Documentation
Piezo Pico-to-Femtotechnology Sensors Inc.
Version 1.0 — August 2026

1. Overview

This documentation presents the API architecture supporting the patented Piezo Pico-to-Femtotechnology Sensors Inc. biosensing device, the VCE-100 machine-learning classification models (**V2, V6, and v2/v6 Compare**), the virus database, and the educational workflow designed for student laboratories. It integrates simulation, classification, noise analysis, virus-range filtering, and model comparison to demonstrate how biosensing data are processed and interpreted using Piezo-Pico to Femtotechnology Sensors Inc. patented technology

2. Device Documentation (Patented Technology)

2.1 Device Description

The patented biosensing device detects frequency noise variations down to 10^{-10} Hz and is trained to identify viral mass at resonant or base frequencies. When a virus binds to the sensor surface, the resonant frequency shifts — this change is converted into biosensing features that feed into trained ML models for classification.

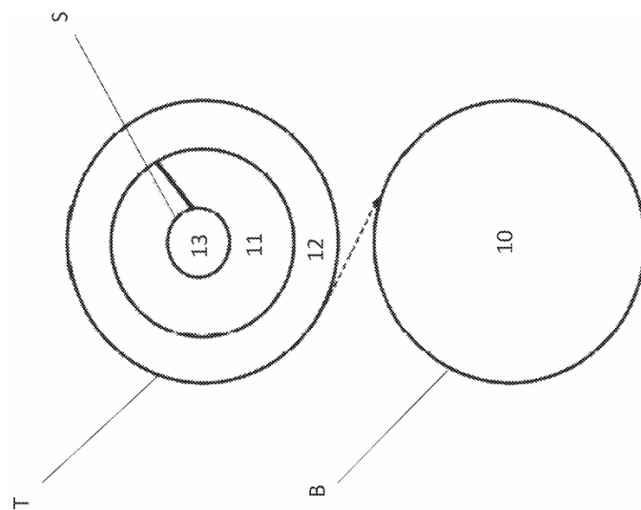
Patented Innovations

- Nano-resonant biosensor
- Frequency-shift detection algorithm
- Mass-estimation pipeline
- Multi-class probability classification
- Real-time simulation mode

Device Architecture

- Sensor Module: AT-Cut QCM mass sensor
- Signal Processor: Converts analog resonance into digital features
- ML Engine: Implements V2, V6, and Spectral models
- API Gateway: FastAPI backend
- Dashboard: React-based frontend interface

Device Images



S = Center dot electrode, T = Quartz disk between top and bottom electrodes, B = Bottom electrode, 10 = Full gold electrode at bottom, 11 = Blank quartz ring, 12 = Gold electrode ring, 13 = Dot center electrode.

3. API Documentation (Swagger & Folder Structure)

The API backend is built with FastAPI, exposing endpoints for simulation and classification. Detailed Swagger specifications and folder structure are documented separately in the following reference files:

- Swagger Documentation: *Swagger.pdf*
- Folder Structure of Selected ML Models:
Folder Structure of Selected ML Models.pdf

3.1 API Architecture

The backend exposes:

- /Simulate — Generates biosensing features
- /Classify/v2 — RandomForest model , XBoost model
- /Classify/v6 — RandomForest model
- /Classify/spectral — Spectral signature model

Swagger UI: <http://127.0.0.1:8000/docs> OpenAPI JSON:
<http://127.0.0.1:8000/openapi.json>

4. Virus Table Documentation

The ML models use a 100-virus database containing:

- Virus ID
- Virus Name
- Mass (fg)
- Transmission Mode
- ML Probability

Virus masses were derived from literature and normalized to sensor response. Access to the table is available after subscription to the ML models for one month.

5. ML Model Documentation

5.1 V2 Model — XBoost (12 000 samples), RandomForest model (12 000 samples)

Training Summary

The V2 classification model was trained using **(12 000 samples** on 12 000 samples, achieving 88.5 % accuracy.

ML Training V2

Execute

Model Name: Analyzer V2

Dataset Size: Small Dataset (12,000 samples)

Status: Training Completed

Accuracy: 88.5%

Loss: 0.31

Last Trained: 2026-06-30

Epochs: 30

Training Time (min): 9

Logs

Epoch 1/30 - loss: 1.12 - acc: 52%

Epoch 10/30 - loss: 0.61 - acc: 71%

Epoch 20/30 - loss: 0.41 - acc: 81%

Epoch 30/30 - loss: 0.31 - acc: 88.5%

Educational Purpose

Students learn:

- Early-stage ML model behavior
- Noise sensitivity
- Why XBoost and cannot identify every virus in the dataset
- How limited datasets affect accuracy

5.2 V6 Model — RandomForest (120 000 samples)

Training Summary

The V6 model was trained using RandomForest on 120 000 samples, achieving 94.2 % accuracy.

ML Training V6

Execute

Model Name: Analyzer V6

Dataset Size: Large Dataset (120,000 samples)

Status: Training Completed

Accuracy: 94.2%

Loss: 0.12

Last Trained: 2026-06-30

Epochs: 50

Training Time (min): 18

Logs

Epoch 1/50 - loss: 0.89 - acc: 62%

Epoch 10/50 - loss: 0.45 - acc: 78%

Epoch 25/50 - loss: 0.22 - acc: 88%

Epoch 50/50 - loss: 0.12 - acc: 94.2%

Educational Purpose

Students learn:

- Ensemble learning principles
- How RandomForest identifies every virus in the trained algorithm
- How larger datasets improve accuracy
- How noise affects classification stability

6. Noise-Based Virus Prediction Tables

Tables 1 and 2(a) illustrate how students can predict viruses at different noise levels.

V2 RandomForest model (12 000 samples) Noise Table

Students use this table to study how noise influences early-stage ML models.

Table 1: Top Prediction of Virus with VCE-100 v2 (XBoost) at Different Frequency Noises

Noise (Hz)	Virus ID	Mass (fg)	Virus Name	Mode of Transmissions
0.1				
0.01				
0.001				
0.0001				
0.00001				
0.000001				
0.0000001				
0.00000001				
0.000000001				
0.0000000001				
0.1234567890				
0.0123567890				
0.0023672980				
0.0004325678				
0.0000236715				
0.0000072345				
0.0000003567				
0.0000000563				
0.0000000053				

By using table 1 above the students will be able to predict the detected virus at different frequency noises using v2 ML model.

V6 RandomForest (120, 000 samples) Noise Table

Students observe that RandomForest remains stable and can identify viruses even at extremely low noise levels.

Table 2 (a): Top Prediction of Virus with VCE-100 v6 (Randomforest) at Different Frequency Noises

Noise (Hz)	Virus ID	Mass (fg)	Virus Name	Mode of Transmissions
0.1				
0.01				
0.001				
0.0001				
0.00001				
0.000001				
0.0000001				
0.00000001				
0.000000001				
0.0000000001				
0.1234567890				
0.0123567890				
0.0023672980				
0.0004325678				
0.0000236715				
0.0000072345				
0.0000003567				
0.0000000563				
0.0000000053				

By using the table 2 (a) above, the student will be able to understand that using more samples in the modeling helps to pinpoint every virus in the stored trained algorithm.

7. Virus-Range Filtering (Table 2 b)

By using Table 2(b), students can filter viruses with the lowest probabilities to identify or predict the detected virus.

This teaches:

- Probability-based filtering
- Range-based virus narrowing
- How ML models behave across subsets of virus IDs

Table 2 (b): Top Prediction of Virus with VCE-100 v6 (Random Forest) at Different Virus ID Range

Noise (Hz)	Virus ID Range	Mass (fg)	Virus Name	Mode of Transmissions
0.1	1-100			
0.01	1-100			
0.001	1-100			
0.0001	1-100			
0.00001	1-100			
0.000001	1-100			
0.0000001	1-100			
0.00000001	1-100			
0.000000001	1-100			
0.0000000001	1-100			
0.00001	5-100			
0.00001	11-100			
0.00001	59-67			
0.000001	1-100			
0.000001	7-34			
0.000001	40-76			
0.000001	58-100			
0.000001	58-78			
0.000001	79-100			

By using table 2b, the student will be able to filter the virus with least probabilities and identify or predict the detected virus.

